

Nonlinear dynamic response analyses of Onshore Wind Turbines with Steel–Concrete Hybrid Tower using a co-simulation approach

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ABSTRACT

The Onshore Wind Turbine (OWT) with a Steel–Concrete Hybrid Tower (SCHT) exhibits unique dynamic characteristics due to its combination of a steel upper section and a concrete lower section, distinguishing it from traditional steel towers. This study proposes a co-simulation approach to develop an integrated, fully-coupled model of OWTs with SCHT, combining multi-body dynamics and finite-element methods for nonlinear dynamic response analysis. First, the multi-flexible body system of the OWT is divided into the blade system and the tower system, with a force–force coupling method applied to manage the interactions between these components. The equation of motion for the blade system is derived using the Euler–Lagrange equations, establishing a multi-body dynamic model. This allows for the detailed simulation of blade dynamics, capturing their flexible behavior and interactions with other turbine components. Then, a finite-element model employing beam elements is developed with OpenSEES software to describe the structural behavior of the SCHT, incorporating material nonlinearity. This model captures the distinct structural responses of the steel and concrete sections. The interactions between these two separate systems are then integrated using an iterative coupling method to formulate the co-simulation approach. This iterative method ensures accurate representation of the dynamic interactions between the two subsystems, leading to a more realistic simulation of the overall turbine behavior. Numerical investigations are conducted to verify the effectiveness of the proposed co-simulation model. Additionally, both linear and nonlinear dynamic response analyses of OWTs with SCHT under extreme wind conditions are performed.

1. Introduction

As the demand for renewable energy continues to surge, improving the efficiency and reliability of wind turbines has become a crucial focus of research and development. Onshore Wind Turbines (OWTs), favored for their ease of installation and lower costs compared to offshore systems, encounter various dynamic loads that can impact their performance and structural integrity. One innovative solution to enhancing the stability and durability of these turbines is the implementation of Steel–Concrete Hybrid Towers (SCHTs). The SCHT combines the strengths of both materials to provide a robust and resilient support structure for wind turbines. This combination enhances overall structural performance, enabling the construction of taller and more robust wind turbines capable of withstanding diverse environmental loads. Additionally, since the concrete section can exhibit nonlinear behaviors, a nonlinear dynamic response analysis of OWTs with SCHTs is crucial for

a comprehensive understanding of their performance. Due to the complexity of OWT with SCHT, the dynamic response analysis of OWT with SCHT still remains a challenging task. Given the similarities between OWTs with SCHT and Monopile Offshore Wind Turbines (MOWTs), the extensive investigations conducted on the dynamic analysis of MOWTs can serve as valuable references. Traditionally, the Multi-Body Dynamic (MBD) model and Finite-Element (FE) model are widely applied for this purpose.

The multi-body dynamics (MBD) approach is frequently employed to model the entire wind turbine system, capturing the interactions between components such as the rotor, drivetrain, and tower [1,2]. This method facilitates the simulation of complex dynamic behaviors and the evaluation of various operational conditions and external loads. By accounting for the flexibility and movements of these components,

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MBD offers detailed insights into the overall performance and reliability of wind turbines under diverse scenarios. For instance, Harun et al. (2017) utilized the MBD approach to simulate wind turbine dynamic deformations, subsequently applying it as a design method for foundations [3]. Sun et al. (2018) developed an analytical model of a MOWT coupled with a pendulum tuned mass damper, modeling interactions between the blades, tower, and damper [4]. Sajeer et al. (2021) investigated the dynamic response of a MOWT using MBD for hydro-servo-elastic analysis, incorporating soil-structure interaction [5]. Xi et al. (2021) designed a monopile for a 10 MW offshore wind turbine specific to an ocean site, developing a new soil-structure interaction framework for the aero-servo-hydro-elastic code FAST, also based on the MBD approach [6]. Similarly, Xie et al. (2022) established a complete multi-body model of a fixed-bottom offshore wind turbine considering pile-soil interaction, performing simulations under multiple external excitations [7]. Other related works utilizing the MBD approach include studies by Li et al. (2017) [8], Zhang et al. (2020) [9], Jahani et al. (2022) [10], Xi et al. (2022) [11], Padron et al. (2022) [12], Steinacker et al. (2022), and Ding et al. (2024) [13]. Recently, Mitra et al. (2023) extended the MBD approach to study the dynamic behavior of OWTs [14]. Despite these advancements, the MBD approach can only simulate the elastic response of wind turbines. Under the extreme loading conditions, the OWT with SCHT may experience nonlinear deformations, which may not be realistically considered by MBD approach.

On the other hand, the FE method stands as a cornerstone in probing the structural behavior and stress distribution within wind turbine components. This approach offers an intricate exploration of mechanical properties and material behaviors, thereby facilitating a thorough assessment of structural integrity and potential failure modes in wind turbines. A plethora of investigations has delved into the dynamic responses of wind turbines under varied loading scenarios, encompassing wind, sea waves, and seismic forces. In streamlining these analyses, researchers commonly assume wind turbines to be in a parked state, with blades represented as a lumped mass atop the tower [15–21]. This simplification, however, overlooks the intricate geometrical intricacies of the blades and their interactions with the tower. Conversely, some studies, like those by Prowell et al. [22], Kjølraug and Kaynia [23], and Santangelo et al. [24], have intricately incorporated blade geometries into their FE models for seismic analyses. Despite this, they still constrain their investigations to the parked condition, neglecting the dynamic effects of blade rotation, which can induce position shifts and stiffness variations. The dynamic responses of contemporary NREL 5 MW wind turbines under combined wind and sea wave loads have been scrutinized numerically using the Abaqus FE code [25]. However, the intricate interactions between blades and tower may not receive due consideration. Indeed, simulating blade operation conditions involving rotation using the FE method presents considerable challenges, accompanied by significant computational efforts.

Given the aforementioned limitations, this study proposes an innovative Co-Simulation (CS) approach for dynamic response analysis of OWT with SCHT. The methodology involves segregating the OWT with SCHT into two distinct systems: the blade system is simulated using the MBD approach, while the FE model is employed to investigate the structural behavior of the SCHT. Additionally, proper consideration is given to the interactions between these two segregated systems. The subsequent sections of the paper are structured as follows: Section 2 delineates the model description of the OWT with SCHT. Section 3 elaborates on the development of the CS model for dynamic response analyses. The verification of the proposed CS method and subsequent numerical analyses are conducted in Section 4. Finally, concluding remarks are provided in the last section.

2. Model description of OWTs with SCHT

This section provides a general description of an OWT with SCHT. Additionally, the parameter settings for the blades and tower are also provided.

2.1. General description

A SCHT wind turbine comprises two primary sections: the upper part, which includes the blades and nacelle components, and the lower part, which is the SCHT. Fig. 1 provides a schematic illustration of the OWT with SCHT configuration. The wind turbine structure is designed to withstand wind loads in this paper. For simplicity, this study assumes the tower base to be a fixed support and considers the nacelle and hub as rigid bodies. In practical engineering applications, the initial section of the tower may contain openings; however, these are typically reinforced along their edges. Therefore, for the purposes of this study, these openings are disregarded in the overall dynamic response analysis of the tower [26].

In Fig. 1, a global coordinate system $OXYZ$ is established with the intersection of the ground and tower axis as the origin O . The Z -axis aligns with the tower axis, the Y -direction parallels the blade pitching, and the X -direction is perpendicular to the blade flapping. The degrees of freedom q_7 and q_8 represent the fore-aft and side-side responses of the nacelle atop the tower relative to its base.

For the blades, as depicted in Fig. 2, a local coordinate system $o'x'y'z'$ is established parallel to the global coordinates $OXYZ$, with the hub center as the origin. In this coordinate system, q_1 , q_2 , and q_3 represent the degrees of freedom for the edgewise (out-of-plane) motions of the three blades within their respective local systems. Similarly, q_4 , q_5 , and q_6 represent the degrees of freedom for the flapwise (in-plane) motions of the three blades within their respective local systems.

In normal operation, wind energy is converted into electrical energy as the blades rotate around the hub center, causing their positions to change over time. If we denote the angular velocity of the blade rotation as Ω , the azimuth angle $\psi_j(t)$ of the j th blade at time instant t can be expressed as:

$$\psi_j(t) = \Omega t + \frac{2\pi}{3}(j - 1), j = 1, 2, 3 \quad (1)$$

2.2. Parameter settings of OWTs with SCHT

In this section, the 10 MW (DTU-10 MW) wind turbine blade model provided by the Technical University of Denmark [27] is selected for the upper blades. The flexible deformation characteristics of the tower and blades are captured using the modal synthesis method. Accordingly, the fundamental mode shapes of the blades and tower are also presented in this section.

2.2.1. Blade parameters

The SCHT turbine blades are modeled after those of the DTU 10 MW turbine. This wind turbine uses a three-blade configuration for power generation, with a rotor diameter of 178.3 m and individual blade lengths of 86.35 m. The total weight of the blades is 125,078.565 kg, while the nacelle weighs 227,962 kg. Key parameters for the upper components are summarized in Table 1.

The fundamental mode functions of the blade in edgewise and flapwise directions, denoted by $\phi_{1e}(r)$ and $\phi_{1f}(r)$ respectively, are defined as follows: [28]

$$\begin{aligned} \phi_{1f}(r) &= -0.62450 \times r^6 + 0.08439 \times r^5 + 1.26100 \times r^4 + 0.14430 \times r^3 \\ &\quad + 0.13510 \times r^2 \\ \phi_{1e}(r) &= -0.06974 \times r^6 + 0.71490 \times r^5 + 0.45620 \times r^4 + 0.82800 \times r^3 \\ &\quad + 0.36200 \times r^2 \end{aligned} \quad (2)$$

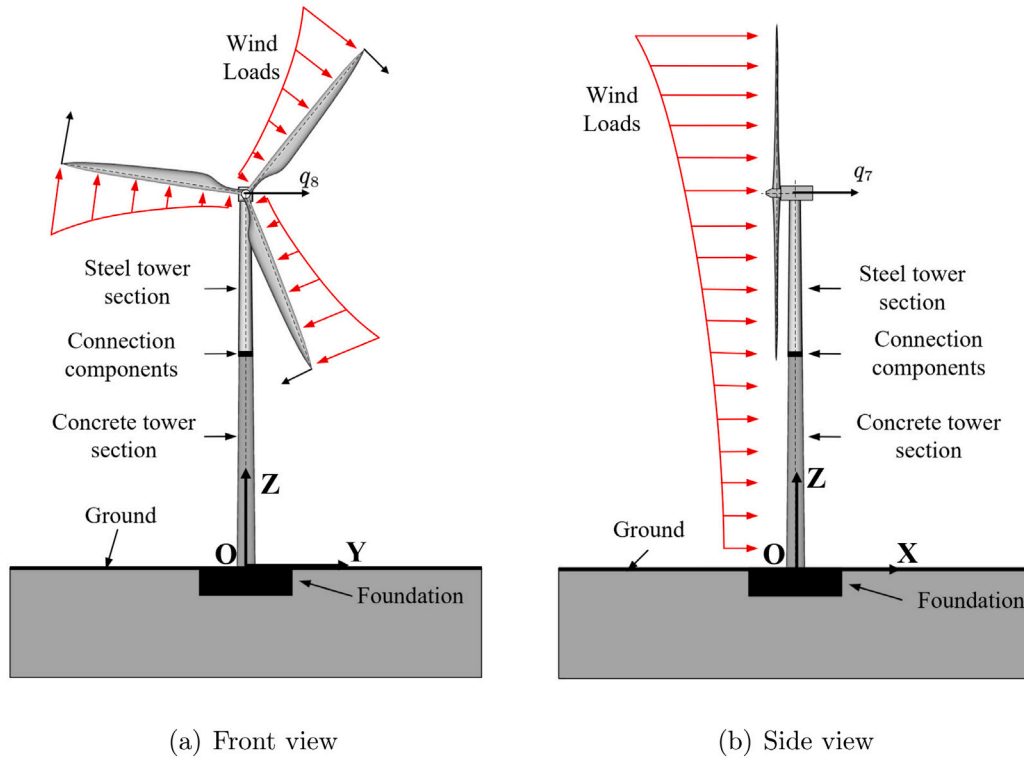


Fig. 1. Schematic illustration of OWTs with SHT.

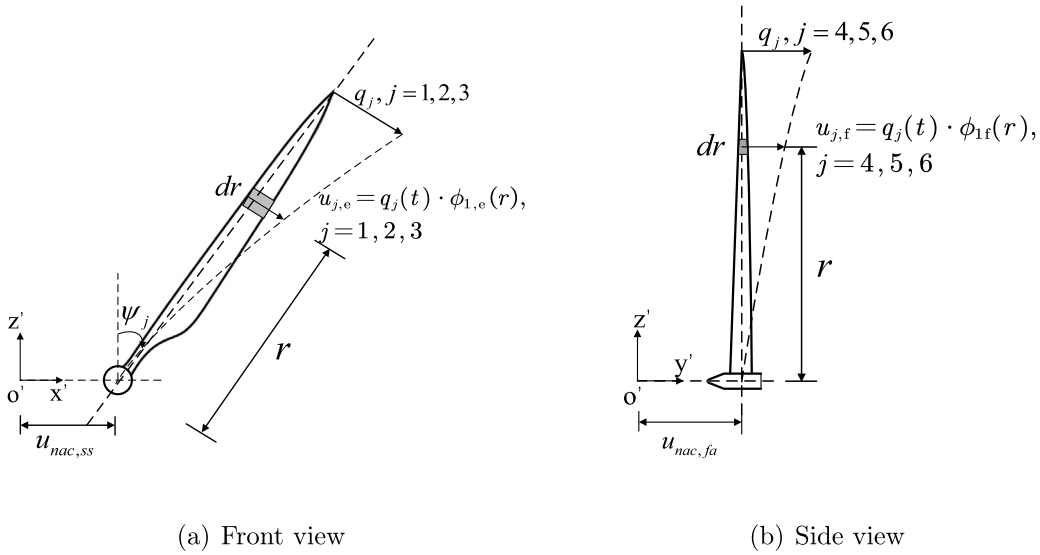


Fig. 2. Schematic illustration of wind turbine blades.

Table 1
Parameters of the DTU 10 MW turbine.

Gross properties	Rating	10 MW
	Rotor orientation, Configuration	Upwind, 3 Blades
	Rotor, Hub diameter	178.3 m, 5.6 m
	Hub height	160.0 m
	Cut-in, Rated, Cut-out wind speed	4 m/s, 11.4 m/s, 25 m/s
	Cut-in, Rated rotor speed	6.0 rpm, 9.6 rpm
Blade structural properties	Length	89.2 m
	Overall (integrated) mass	416,92.855 kg
	Structural-damping ratio	0.48%
Nacelle and hub properties	Hub mass	105,520 kg
	Nacelle mass	446,006 kg

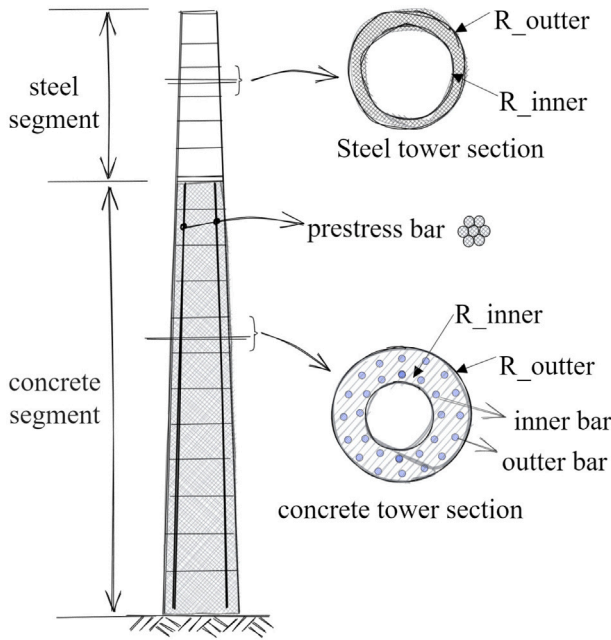


Fig. 3. Simplified diagram of SCHAT model.

Table 2
Material parameters.

Material	Parameter	Value
C70	Compressive strength	44.5 MPa
	Tensile strength	2.99 MPa
	Elastic modulus	37 000 MPa
C65	Compressive strength	41.5 MPa
	Tensile strength	2.93 MPa
	Elastic modulus	36 500 MPa
HRB335	Yield strength	335 MPa
	Ultimate strength	455 MPa
	Elastic modulus	200 000 MPa
Prestressed tendon	Yield strength	1320 MPa
	Ultimate strength	1860 MPa
	Elastic modulus	195 000 MPa
Q345	Yield strength	345 MPa
	Ultimate strength	470 MPa
	Elastic modulus	206 000 MPa

2.2.2. Tower parameters

As illustrated in Fig. 3, the SCHAT is a steel–concrete hybrid tower consisting of a steel upper section and a concrete lower section, with a total height of 158 m. The concrete section extends from 0 m to 112 m, while the remaining portion is composed of steel. The tower of the SCHAT wind turbine is modeled as a typical cantilever beam structure, with its cross-sectional geometric characteristics varying along the height of the tower.

For the concrete tower, the lower two sections utilize C70 concrete as per Chinese standards, while the remaining sections use C65 concrete. The reinforcement for the concrete sections employs HRB335 steel bars. Additionally, the prestressed tendons are made of externally unbonded, high-strength, low-relaxation wire ropes with a diameter of 15.2 mm and an initial prestress of 1280 MPa. For the steel sections of the tower, Q345 steel is used. Detailed material parameters are provided in Table 2.

Then, the first-mode vibration function of the tower, denoted as $\phi_{1t}(z)$, can be expressed as [29]:

$$\phi_{1t}(z) = 0.36850 \times z^6 - 2.55828 \times z^5 + 3.47648 \times z^4 - 1.87776 \times z^3 + 1.59106 \times z^2 \quad (3)$$

2.3. Wind loads

During operation, an OWT with SCHAT relies on its blades to harness wind energy, bearing the majority of the wind loads and transferring these loads downwards. This section will delve into the theoretical calculations of the wind loads experienced by an OWT with SCHAT. As is commonly understood, wind speed comprises both mean and turbulent components [30], denoted by $v(t) = \bar{v} + \tilde{v}(t)$. The mean wind speed at a given height z is determined through a power-law profile, whereas the turbulent component of wind speed is calculated using the spectral representation method. For the latter, the Kaimal spectrum is utilized as the reference spectrum [31].

As illustrated in Fig. 4(a), the program calculates the wind speed at each point in a 31×31 grid. This generates a stochastic wind field, which is then applied to the blades and tower structures to assess the effects of turbulent wind-induced loadings on the wind turbine. Fig. 4(b) depicts the time history of wind speed at a specific point in the turbulent wind field.

The Blade Element Momentum (BEM) theory [28,32,33] is then employed to approximate the load effects acting on the blades. This method offers high computational efficiency compared to computational fluid dynamics, while also maintaining a high level of accuracy. BEM theory combines blade element theory and momentum theory to determine the normal forces P_N and tangential forces P_T acting on blade elements, based on the aerodynamic characteristics of the blades and the incoming wind speed profiles. Recognizing that direct iteration methods exhibit poor stability, this paper adopts the computational approach outlined in [34,35] to transform the traditional two-variable iterative BEM equations into single-variable equations with unknown inflow angles, employing stable and reliable root-finding algorithms to solve them. After obtaining the aerodynamic loads, specifically the normal forces P_N and tangential forces P_T , through the BEM theory, the generalized aerodynamic loads acting on each degree of freedom are calculated using the principle of virtual work. The virtual work generated by the aerodynamic loads is expressed as follows [33]:

$$\delta W_{wl} = \sum_{j=1}^3 \left\{ \int_0^R P_{Tj}(r, t) [\phi_{1c}(r) \delta q_j + \delta u_{nac}^{ss} \cos \psi_j] dr + \int_0^R P_{Nj}(r, t) [\phi_{1f}(r) \delta q_{j+3} + \delta u_{nac}^{fa}] dr \right\} \quad (4)$$

where $P_{Tj}(r, t)$ and $P_{Nj}(r, t)$ respectively represent the tangential and normal forces experienced by the j th blade element.

Introducing the generalized force Q_j , in accordance with the relationship between work and energy, the virtual work done by the generalized force is given by:

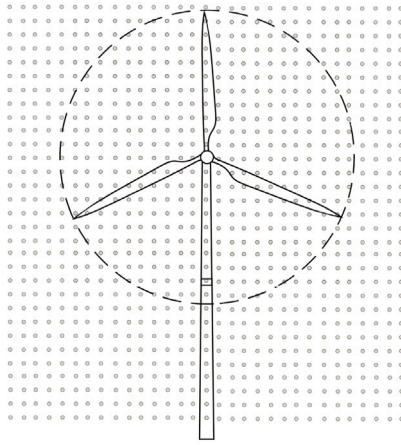
$$Q_j = \frac{\partial W_{wl}}{\partial q_j} \quad (5)$$

Substituting Eq. (4) into Eq. (5) yields the generalized forces acting on each degree of freedom as follows:

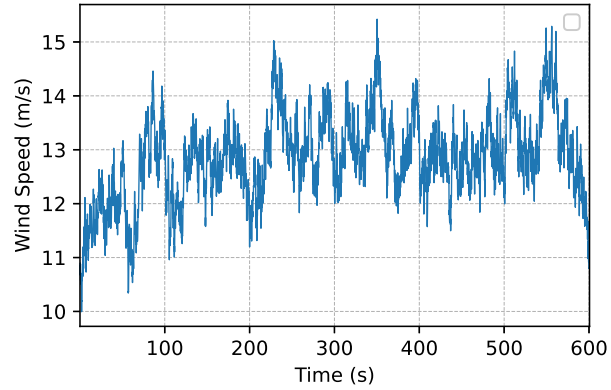
$$Q_{j,wind} = \int_0^R P_{Tj}(r, t) \phi_{1c} dr, \quad Q_{j+3,wind} = \int_0^R P_{Nj}(r, t) \phi_{1f} dr, \quad j = 1, 2, 3$$

$$Q_{7,wind} = \sum_{j=1}^3 \int_0^R P_{Tj}(r, t) dr, \quad Q_{8,wind} = \sum_{j=1}^3 \int_0^R P_{Nj}(r, t) dr \cos \psi_j \quad (6)$$

This section has provided a comprehensive overview of the OWT with SCHAT, detailing the parameters for both the blades and tower materials. Additionally, the wind loads acting on the turbine have been described. With this foundational information, the subsequent section will introduce a CS approach to perform nonlinear dynamic



(a) Wind speed grid



(b) Typical time history of wind speed

Fig. 4. Wind speed.

response analysis of OWTs with SCHAT. This approach will integrate multi-body dynamics and FE methods to provide a detailed and accurate assessment of the turbine's behavior under various loading conditions.

3. Co-simulation model for wind turbines with steel-concrete hybrid tower

In a multi-body system, different components often exhibit varying levels of structural complexity. Employing a uniform analysis method across the entire system can result in inaccurate results or high computational costs. Therefore, it is advantageous to decompose structures of varying complexities into relatively independent components. These components can be analyzed separately and their individual solutions integrated afterward, reducing computational costs and enhancing accuracy. This method is referred to as the CS approach [36]. Considering the complexity of the SCHAT section and its material behavior, along with the relatively simple characteristics of the blades and hub, a co-simulation model will be established to develop an integrated, fully-coupled model for the SCHAT wind turbine.

3.1. Introduction of co-simulation approach

For multi-body systems involving flexible bodies, a Co-Simulation (CS) approach that combines the MBD method with the FE model can be employed. This approach allows for the creation of a model for the flexible body within an independent coordinate system, accurately describing its elastic deformation. In practice, simulations of multi-flexible body systems have been explored across various fields, such as vehicle-track coupling systems [36] and gear box systems of wind turbines [37,38], among others. Typically, this involves integrating commercial simulation software like SIMPACK and Abaqus to perform hybrid simulations [39–42]. This combined approach leverages the strengths of both MBD and FE methods, enabling a comprehensive analysis of the dynamic responses and interactions within flexible multi-body systems.

As illustrated in Fig. 5, a multi-flexible body system M is coupled with the subsystem S through a fixed connection. The multi-flexible body system M also can comprise multiple flexible bodies S_{bi} and stiffness components, which can be solved as a complete multi-flexible body system [43]. The interaction between the multi-flexible body system M and the subsystem S can be likened to a black box, producing different output results under varying input conditions such that

$$\begin{cases} x_s = S(\bar{\lambda}_c) \\ x_m = M(\bar{\lambda}_c) \end{cases} \quad (7)$$

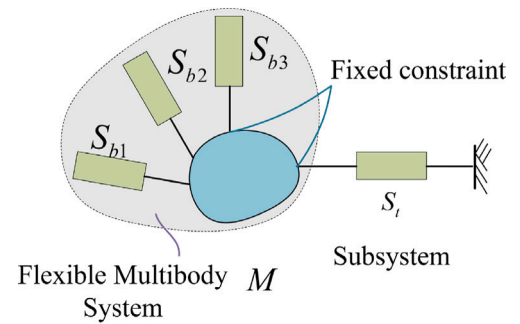


Fig. 5. Simplified diagram of CS model.

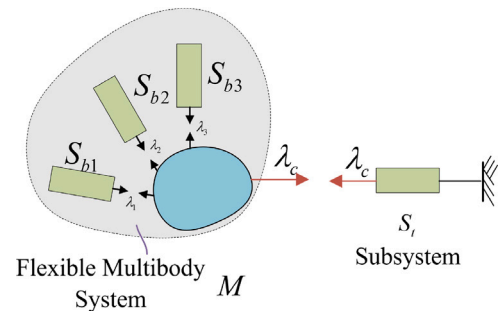


Fig. 6. CS model: force-force coupling.

where x_s and x_m denote the responses of the subsystem S and the multi-flexible body system M , respectively. These responses can encompass displacement, velocity, or acceleration as structural response variables. Meanwhile, $\bar{\lambda}_c$ represents the generalized force acting on both systems.

As depicted in Fig. 6 and based on Eq. (7), the two systems can be decoupled using a force-force coupling method. At each macro time step, which corresponds to the interval for data exchange between the systems, the response variables of each system are computed. These response variables are then used to calculate the generalized loads that act as inputs to the other system for the subsequent time step. The force-force coupling method is well-suited for a wide range of applications, as it enables the decomposition of complex systems into subsystems, allowing for parallel computation and simplified interaction modeling through constitutive laws. Each subsystem can be

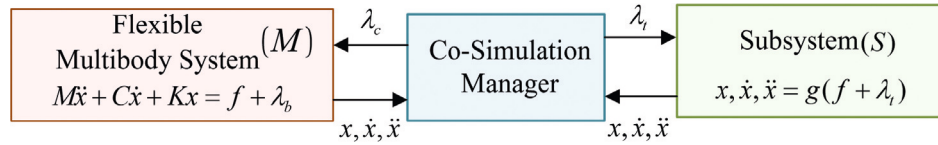


Fig. 7. Diagram of the CS model.

solved independently using solvers tailored to its specific requirements. However, the stability and accuracy of the force–force coupling method are sensitive to coupling parameters, and its non-iterative implementation may suffer from reduced stability, particularly in systems with strong coupling or high stiffness. When combined with iterative coupling methods, it achieves excellent numerical stability and effectively handles such challenges, albeit at the cost of increased computational effort [44]. Through an iterative process of data exchange at macro time steps, the interaction forces between the subsystems can be accurately captured and updated.

As shown in Fig. 7, for a multi-body system, the equations of motion can be derived using methods such as Euler–Lagrange equations and solved using algorithms like Runge–Kutta or Newmark. Meanwhile, the structural response of the subsystem can be determined by solving the implicit function $g(\cdot)$. These two systems transfer structural displacement, velocity, and acceleration to the CS manager, which then calculates the generalized forces acting on the other system. When the computer performance is strong enough or the interaction cost between the two systems is low, using smaller macro time steps can reduce errors. However, in mature and stable time integration algorithms, adaptive time steps are commonly employed to enhance solution stability and reduce computation time, making it challenging to select a suitable macro time step. Therefore, some studies used extrapolation methods to replace actual responses in other systems [36,44]. For simplification purposes, this paper will utilize linear interpolation to calculate the response outputs of each subsystem towards the other systems within their respective sub-time steps, and perform generalized force calculations for the model in Fig. 7.

For dynamic response analysis of an OWT with SCHT, the flexible multi-body system comprises the blades and nacelle, while the subsystem is treated as the SCHT. In this setup, the flexible multi-body system and the subsystem interact through the CS manager, which facilitates the exchange of structural responses between the two. By doing so, the CS manager calculates the generalized forces acting on each component, ensuring an accurate representation of the overall dynamic behavior of an OWT with SCHT.

3.2. Modeling of flexible multi-body system: Blades

As mentioned, the dynamic responses of blades are evaluated using the equation of motion for the blade system, which is derived from the Euler–Lagrange method and solved with appropriate numerical algorithms such as the Runge–Kutta or Newmark algorithm. These responses include displacements, velocities, and accelerations, which are then communicated to the CS manager.

In our previous work, an 8-DOF multi-body dynamic model has been thoroughly derived for the OWT with SCHT based on the Euler–Lagrange equations [29], which can be expressed as

$$\tilde{M}\ddot{q} + \tilde{C}\dot{q} + \tilde{K}q = \tilde{Q} \quad (8)$$

where $\tilde{M}$, $\tilde{C}$, $\tilde{K}$ denote the mass, damping, and stiffness matrices of the system, respectively. The term $\tilde{Q}$ represents the system's generalized force vector. Detailed explanations of the system matrices are provided in Appendix.

In this context, the equation of motion for the blade system can be expressed in a general form as follows:

$$M_{\text{blade}} \cdot \ddot{q}_{\text{blade}} + C_{\text{blade}} \cdot \dot{q}_{\text{blade}} + K_{\text{blade}} \cdot q_{\text{blade}} = Q_{\text{blade}} \quad (9)$$

where M_{blade} , C_{blade} , K_{blade} denote the mass, damping, and stiffness matrices for the blade system, respectively, q_{blade} , $\dot{q}_{\text{blade}}$, $\ddot{q}_{\text{blade}}$ represent the displacement, velocity, and acceleration vectors of the blade system, and the term Q_{blade} represents the blade system's generalized force vector. Actually, these matrices can be considered as the sub-matrices that correspond to the first six degrees of freedom in the matrices $\tilde{M}$, $\tilde{C}$, $\tilde{K}$.

3.3. Modeling of subsystem: SCHT model

As shown in Fig. 3, the upper part of the SCHT is made of a steel tower, while the lower part consists of a reinforced concrete tower body. The reinforced concrete section contains prestressed steel reinforcements to induce compressive stress between tower segments. Additionally, the tower features a reinforcing steel mesh, with different segment configurations for both reinforcement and geometry. To accurately capture the structural behaviors of the SCHT, a nonlinear FE model needs to be established and applied. For this purpose, the OpenSEES software is specifically employed.

The SCHT model in OpenSEES consists of 31 sections of reinforced concrete tower segments and 4 sections of steel tower segments, with each tower represented by one beam element, as depicted in Fig. 8. The base of the tower is fixedly constrained, while the wind turbine hub loads at the top are calculated using the MBD model built in the previous section. However, the MBD model cannot account for the vertical direction (Z -axis) of the tower. Therefore, in the OpenSEES model, a lumped mass in the Z -axis is added to simulate the hub loads.

Both linear and nonlinear models will be established using displacement-based beam–column elements, with unit sections defined in a fiber manner as depicted in Fig. 9. For reinforced concrete tower segments, longitudinal steel fibers will be created alongside concrete section fibers, as shown in Fig. 10. In this representation, steel fibers are arranged based on the actual quantities and positions of longitudinal reinforcement. Since simulating reinforcement stirrups is challenging in OpenSEES, an approximate modeling approach using the Mander confined concrete model is adopted to simulate the strengthening effect of stirrups and longitudinal reinforcements on concrete. The same fiber approach is applied to steel tower segments.

In the nonlinear models, both material and geometric nonlinearities are incorporated within OpenSEES. For material nonlinearity, the “Concrete D” material model is used to simulate the nonlinear behavior of concrete fibers [45], while steel fibers and steel tower sections are modeled with the bilinear “Steel01” material model. Prestressed steel bars, subjected to uniaxial tension only, are represented using truss elements. Geometric nonlinearity of the P - Δ effect is incorporated via the “geomTransf PDelta” transformation, capturing significant tower deformations under dynamic loading conditions. As shown in Fig. 10, to simultaneously observe the stress–strain response of linear and nonlinear hybrid models and to reduce output burden, eight stress–strain output monitoring fibers were set along the circumference every 45° on the outer layer of concrete fiber in the lower portion of the tower. This placement is based on the fact that the maximum stress–strain values generally occur at the bottom of the tower. The monitoring fibers are named clockwise starting from $x = 0, y > 0$ as “Node1”, “Node2”, up to “Node8”.



Fig. 8. OpenSEES model.

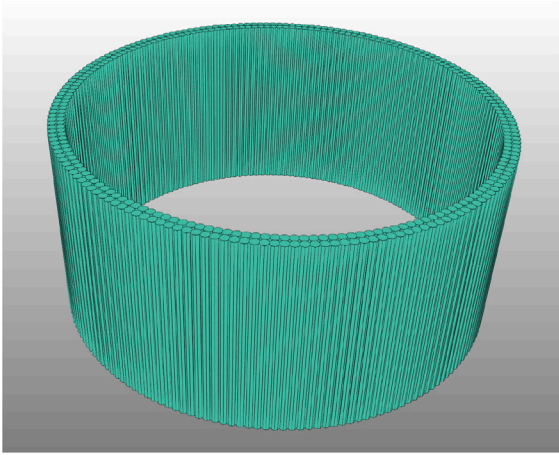


Fig. 9. First tower section in OpenSEES.

3.4. Development of the co-simulation model for wind turbine with SCHT

As mentioned, an OWT with SCHT can be modeled as a multi-body system M , consisting of three flexible blade bodies S_{b1}, S_{b2}, S_{b3} and the rigid body of the nacelle. The subsystem S represents the tower model, which is established using the FE method. The dynamic response of the rotor blades interacts with that of the tower. As shown in Appendix A.3 and Eq. (8), the model actually combines the six degrees of freedom of the wind turbine blades with the two degrees of freedom of the lower support structure. Therefore, it can be expanded as follows:

$$\begin{bmatrix} \mathbf{M}_{\text{blade}} & \mathbf{M}_{\text{side,up}} \\ -\mathbf{M}_{\text{side,down}} & \mathbf{M}_{\text{tower}} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}}_{\text{blade}} \\ \ddot{\mathbf{q}}_{\text{tower}} \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\text{blade}} & \mathbf{C}_{\text{side,up}} \\ -\mathbf{C}_{\text{side,down}} & \mathbf{C}_{\text{tower}} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{q}}_{\text{blade}} \\ \dot{\mathbf{q}}_{\text{tower}} \end{bmatrix} + \begin{bmatrix} \mathbf{K}_{\text{blade}} & \mathbf{K}_{\text{side,up}} \\ -\mathbf{K}_{\text{side,down}} & \mathbf{K}_{\text{tower}} \end{bmatrix} \begin{bmatrix} \mathbf{q}_{\text{blade}} \\ \mathbf{q}_{\text{tower}} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_{\text{blade}} \\ \mathbf{Q}_{\text{tower}} \end{bmatrix} \quad (10)$$

where $\mathbf{M}_{\text{tower}}, \mathbf{C}_{\text{tower}}, \mathbf{K}_{\text{tower}}$ denote the mass, damping, and stiffness matrices of the top two degrees of freedom for the tower; $\mathbf{q}_{\text{tower}}, \dot{\mathbf{q}}_{\text{tower}}, \ddot{\mathbf{q}}_{\text{tower}}$ represent the displacement, velocity, and acceleration vectors of the tower, and $\mathbf{Q}_{\text{tower}}$ is the generalized force of the environmental

loads acting on the tower. Due to the interaction between the upper and lower structures, these matrices are not diagonal but contain nonzero terms in both the upper and lower triangles, which are denoted in $\mathbf{M}_{\text{side,up}}, \mathbf{M}_{\text{side,down}}, \mathbf{C}_{\text{side,up}}, \mathbf{C}_{\text{side,down}}, \mathbf{K}_{\text{side,up}}, \mathbf{K}_{\text{side,down}}$. Additionally, because of the rotation of the rotor, the matrices exhibit characteristics of asymmetric matrices.

Then, Eq. (10) can be further expanded as:

$$\begin{cases} \mathbf{M}_{\text{blade}} \cdot \ddot{\mathbf{q}}_{\text{blade}} + \mathbf{C}_{\text{blade}} \cdot \dot{\mathbf{q}}_{\text{blade}} + \mathbf{K}_{\text{blade}} \cdot \mathbf{q}_{\text{blade}} \\ = -\mathbf{M}_{\text{side,up}} \cdot \ddot{\mathbf{q}}_{\text{tower}} - \mathbf{C}_{\text{side,up}} \cdot \dot{\mathbf{q}}_{\text{tower}} - \mathbf{K}_{\text{side,up}} \cdot \mathbf{q}_{\text{tower}} + \mathbf{Q}_{\text{blade}} \\ \mathbf{M}_{\text{tower}} \cdot \ddot{\mathbf{q}}_{\text{tower}} + \mathbf{C}_{\text{tower}} \cdot \dot{\mathbf{q}}_{\text{tower}} + \mathbf{K}_{\text{tower}} \cdot \mathbf{q}_{\text{tower}} \\ = -\mathbf{M}_{\text{side,down}} \cdot \ddot{\mathbf{q}}_{\text{blade}} - \mathbf{C}_{\text{side,down}} \cdot \dot{\mathbf{q}}_{\text{blade}} - \mathbf{K}_{\text{side,down}} \cdot \mathbf{q}_{\text{blade}} + \mathbf{Q}_{\text{tower}} \end{cases} \quad (11)$$

Assuming $\mathbf{f}_{\text{blade}}$ and $\mathbf{f}_{\text{tower}}$ represent the external force vectors acting on the separated components of a wind turbine's blade system and supporting structure, one can express the equations of motion for both parts as follows:

For the blade system:

$$\mathbf{M}_{\text{blade}} \ddot{\mathbf{q}}_{\text{blade}} + \mathbf{C}_{\text{blade}} \dot{\mathbf{q}}_{\text{blade}} + \mathbf{K}_{\text{blade}} \mathbf{q}_{\text{blade}} = \mathbf{f}_{\text{blade}} \quad (12)$$

For the tower system:

$$\mathbf{M}_{\text{tower}} \ddot{\mathbf{q}}_{\text{tower}} + \mathbf{C}_{\text{tower}} \dot{\mathbf{q}}_{\text{tower}} + \mathbf{K}_{\text{tower}} \mathbf{q}_{\text{tower}} = \mathbf{f}_{\text{tower}} \quad (13)$$

These equations actually account for the dynamics of each component independently, with external forces $\mathbf{f}_{\text{blade}}$ and $\mathbf{f}_{\text{tower}}$ acting on them. The generalized forces $\mathbf{f}_{\text{blade}}$ and $\mathbf{f}_{\text{tower}}$ can be expressed as follows:

$$\begin{cases} \mathbf{f}_{\text{blade}} = \bar{\lambda}_{\text{c,blade}} + \mathbf{Q}_{\text{blade}} \\ = -\mathbf{M}_{\text{side,up}} \cdot \ddot{\mathbf{q}}_{\text{tower}} - \mathbf{C}_{\text{side,up}} \cdot \dot{\mathbf{q}}_{\text{tower}} - \mathbf{K}_{\text{side,up}} \cdot \mathbf{q}_{\text{tower}} + \mathbf{Q}_{\text{blade}} \\ \mathbf{f}_{\text{tower}} = \bar{\lambda}_{\text{c,tower}} + \mathbf{Q}_{\text{tower}} \\ = -\mathbf{M}_{\text{side,down}} \cdot \ddot{\mathbf{q}}_{\text{blade}} - \mathbf{C}_{\text{side,down}} \cdot \dot{\mathbf{q}}_{\text{blade}} - \mathbf{K}_{\text{side,down}} \cdot \mathbf{q}_{\text{blade}} \\ + \mathbf{Q}_{\text{tower}} \end{cases} \quad (14)$$

The equations above demonstrate that both the upper and lower parts of an OWT with SCHT can be described using standard equations of motion. The Eq. (12) is effectively analogous to Eq. (9), indicating that the equation of motion for the blade system can be derived from the 8-DOF model for the OWT with SCHT. Despite the presence of flexible bodies in the upper structure, their deformation characteristics can still be captured using modal synthesis. This approach shows no fundamental deviation from the dynamic models discussed in previous research. Consequently, employing the Runge–Kutta solver can ensure the computational efficiency. For the lower support structure, due to its complex structural characteristics, it is necessary to use the FE method to develop a detailed and sophisticated nonlinear tower model for analysis, where the OpenSEES solver is specifically employed. Therefore, Eq. (12) and (13) can be rewritten as:

$$\begin{cases} \mathbf{q}_{\text{blade}}, \dot{\mathbf{q}}_{\text{blade}} = \text{RK45}(\bar{\lambda}_{\text{c,blade}} + \mathbf{Q}_{\text{blade}}) \\ \ddot{\mathbf{q}}_{\text{blade}} = (\bar{\lambda}_{\text{c,blade}} + \mathbf{Q}_{\text{blade}} - \mathbf{C}_{\text{blade}} \cdot \dot{\mathbf{q}}_{\text{blade}} - \mathbf{K}_{\text{blade}} \cdot \mathbf{q}_{\text{blade}}) \mathbf{M}_{\text{blade}}^{-1} \\ \mathbf{q}_{\text{tower}}, \dot{\mathbf{q}}_{\text{tower}}, \ddot{\mathbf{q}}_{\text{tower}} = \text{OpenSEES}(\bar{\lambda}_{\text{c,tower}} + \mathbf{Q}_{\text{tower}}) \end{cases} \quad (15)$$

where RK45 and OpenSEES represent the Runge–Kutta and OpenSEES solvers, respectively. Specifically, the Runge–Kutta solver is used to solve the equation of motion for the blade system in Eq. (12), while the OpenSEES solver is employed to solve the equation of motion for the tower system in Eq. (13). The generalized force $\bar{\lambda}_{\text{c,blade}}$ corresponds to the force vector exerted by the tower structure subsystem's motion on the upper structure subsystem, whereas $\bar{\lambda}_{\text{c,tower}}$ represents the force vector exerted by the upper structure subsystem's motion on the tower

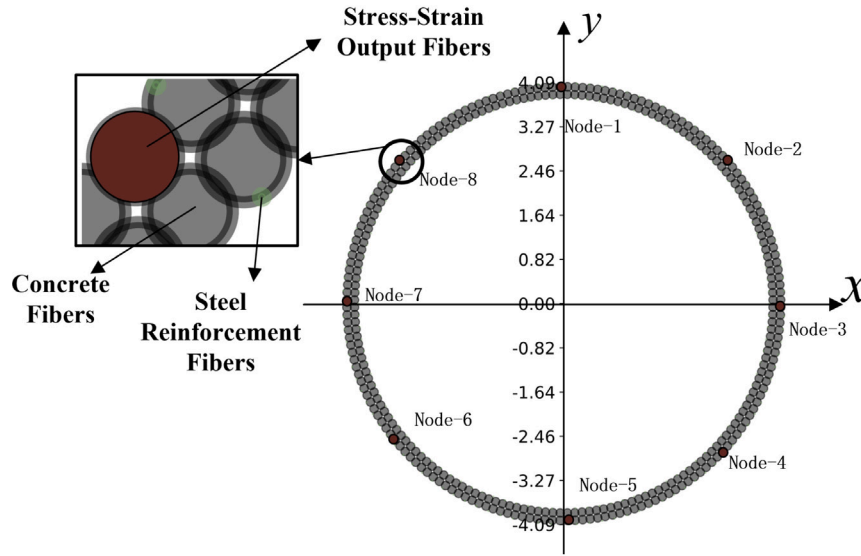


Fig. 10. Cross-sectional view of OpenSEES model.

structure subsystem. According to Eq. (10), the external loads acting on the upper structure and the lower supporting structure are denoted as $\mathbf{Q}_{\text{blade}}$ and $\mathbf{Q}_{\text{tower}}$, respectively.

Substituting Eq. (15) into Eq. (14) yields the iterative equation for calculating the generalized forces as follows:

$$\begin{cases} \tilde{\lambda}_{c,tower} = \mathbf{f}_{tower}(\tilde{\lambda}_{c,blade}, \mathbf{Q}_{blade}) \\ \tilde{\lambda}_{c,blade} = \mathbf{f}_{blade}(\tilde{\lambda}_{c,tower}, \mathbf{Q}_{tower}) \end{cases} \quad (16)$$

As shown in Fig. 11, although the blade and tower systems are separated, Eq. (15) reveals that the dynamic response of the blade system is included in the external force calculation of the tower system, and vice versa. Consequently, there remains an interplay between the rotor system and the tower system, which is a classic problem encountered when dealing with multiple coupled systems and their interactions. In this case, there are two common approaches to simulation: iterative coupling and non-iterative coupling.

Non-iterative coupling involves calculating the external forces acting on each system at a given time step using the responses from the previous time steps for both systems. This approach avoids the need to iterate within the system itself, thereby reducing the control requirements for the solver. However, this method is sensitive to the selection of the time step: if the time step is too small, the computation time and interaction time between systems will increase; if it is too large, the simulation may not converge. Iterative coupling, on the other hand, involves iterating at each time step to achieve equilibrium between the interacting nodes in the two systems. In general, non-iterative coupling requires a more careful selection of the time step and has poorer stability compared to iterative coupling [44,46]. Therefore, this section will employ an iterative approach to simulate the response of the wind turbine system at each time step for error control.

According to Eq. (16), the generalized forces λ_c between systems can be represented by the state of the upper structure and the state of the lower supporting structure. Therefore, it can be utilized as an iterative parameter for calculation [47]. The specific iterative process is illustrated in Fig. 12. This process involves initializing the solver before computational solving begins and selecting initial values for the displacements, velocities, and accelerations of each system's degrees of freedom.

Wind loads on the blades are determined using BEM theory under actual wind field conditions. Utilizing the principle of virtual work, the generalized external forces for every degree of freedom in the wind turbine system are computed. Then, Eq. (16) is used to solve for the external forces on the rotor and tower systems. The respective

solvers are used to determine the displacement and velocity responses of the rotor and tower systems' degrees of freedom. Subsequently, the newly calculated external forces for the rotor and tower systems are substituted back into Eq. (16) and compared with the external forces from the previous iteration step to complete one iteration.

As shown in Fig. 12, in the actual iterative solving process, each time step is decomposed into multiple sub-steps for iteration to reduce the difficulty of solving caused by sudden shifts in system's response. To enhance convergence during iteration, the system's response from the previous time step is utilized as the initial condition for the current time step. This optimization reduces the likelihood of initial value deviation from the actual solution and minimizes the number of iterations required. Concurrently, utilizing the 8-DOF dynamic model established in previous research, a correction model is developed for synchronized solution. This correction model generates system responses that serve as improved initial values for each iteration, further optimizing the iterative process.

Due to the introduction of material nonlinearity and other properties, the stiffness of the tower system may change, leading to discrepancies between the tower body in the correction model and the tower body in the CS model, thereby causing errors in calculations and rendering the selection of initial values meaningless. To address this issue, the tower stiffness values are recalculated using the FE solver OpenSEES at the end of each time step and updated in the correction model to maintain consistency and accuracy across iterations. The iterative error tolerance (*tol*) is set to 10^{-4} , ensuring numerical precision and preventing divergence. To mitigate abrupt changes in system response, linear interpolation is applied within the sub-steps of each time step, enhancing iterative stability and accelerating convergence in solving coupled nonlinear dynamic systems. The iterative procedure for each time step is explained as follows:

- Step 1:** Initialize the solver. Calculate the responses of the 8-DOF dynamic model based on the previous responses, as described in Eq. (A.15), and use these as the initial values $q_{\text{blade}}, \dot{q}_{\text{blade}}, \ddot{q}_{\text{blade}}$ and $q_{\text{tower}}, \dot{q}_{\text{tower}}, \ddot{q}_{\text{tower}}$ for the CS model.
- Step 2:** Calculate the wind loads on the blades using BEM theory under actual wind field conditions at the current time step. Then, compute the generalized external forces for each degree of freedom of the OWT using the principle of virtual work, as described in Eq. (6).
- Step 3:** Calculate the external forces $\tilde{\lambda}_{c,blade}$ acting on the blade system using the first equation in Eq. (14), and solve for the responses

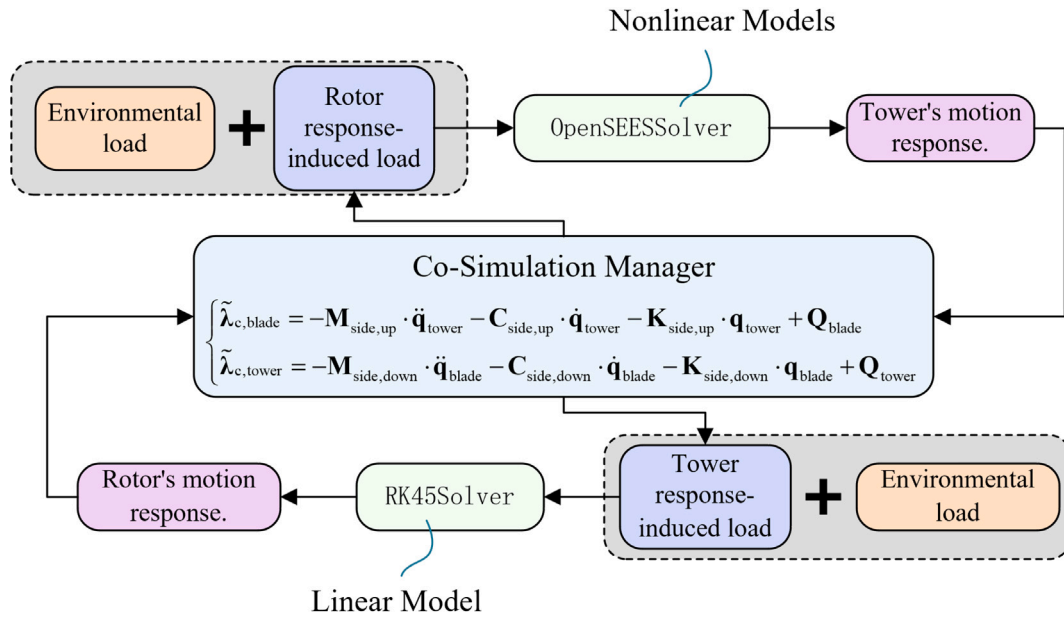


Fig. 11. CS model architecture for an OWT with SCHT.

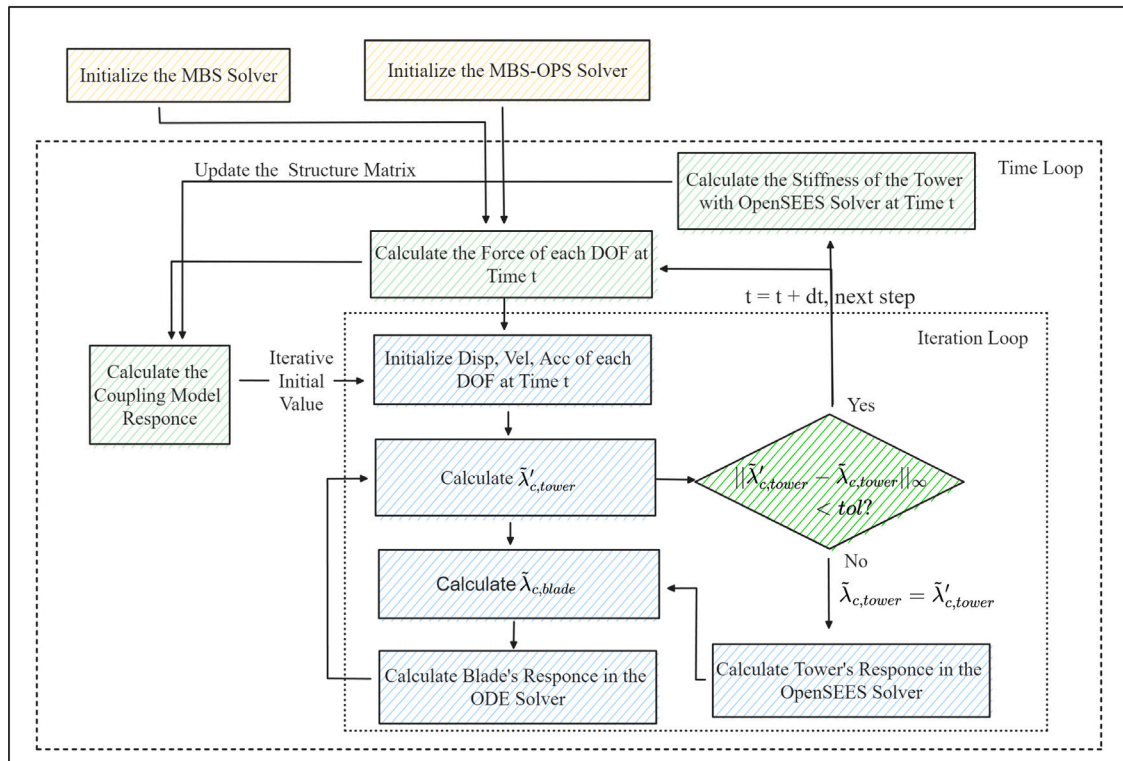


Fig. 12. Flowchart of the iterative process for generalized forces.

of the blade system using the RK45 solver based the equations of motion. The outputs include q_{blade} , $\dot{q}_{blade}$, $\ddot{q}_{blade}$.

Step 4: Calculate the external forces $\tilde{\lambda}_{c,tower}$ acting on the tower system using the second equation in Eq. (14),

Step 5: Iteratively update the load. Compare the current load $\tilde{\lambda}'_{c,tower}$ with the previous load $\tilde{\lambda}_{c,tower}$ using $\|\tilde{\lambda}'_{c,tower} - \tilde{\lambda}_{c,tower}\|_{\infty}$. If the difference is within the set tolerance ($tol = 10^{-4}$), proceed to step 7 and 8; otherwise, proceed to step 6.

Step 6: Use the OpenSEES solver to compute the tower system responses, yielding q_{tower} , $\dot{q}_{tower}$, $\ddot{q}_{tower}$, then return to Step 3.

Step 7: Update the 8-DOF dynamic model matrix $\tilde{\mathbf{M}}$ with tower stiffness values obtained from OpenSEES.

Step 8: Advance the time step $t = t + dt$. If t remains below the target simulation time, return to Step 1; otherwise, terminate the simulation.

It is well known that the OpenSEES solver is an open-source FE computational framework written in C++ [48–50]. To facilitate data interaction between the MBD model and the FE model, as well as to minimize associated costs, this study develops an interaction module for handling data exchange in CS models by conducting secondary

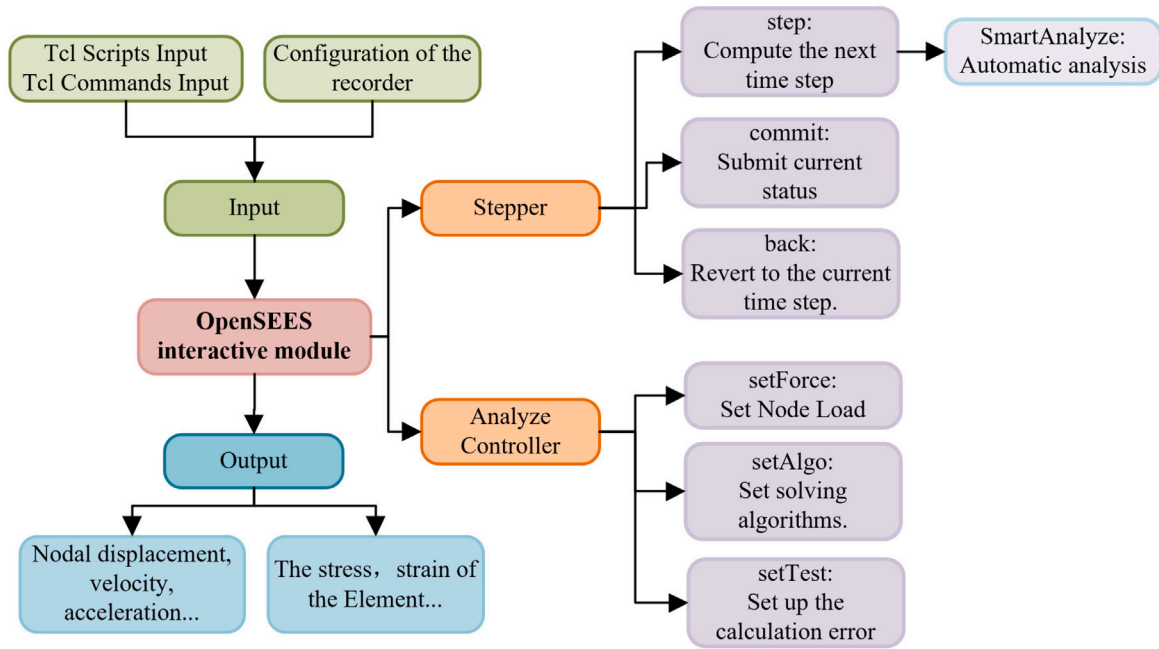


Fig. 13. OpenSEES interactive module.

development of OpenSEES solver. The architecture of the OpenSEES interactive module is illustrated in Fig. 13. The module is responsible for receiving data from the MBD model, calculating the external forces acting on the tower system, and transmitting the results back to the MBD model. Additionally, the module calculates the stiffness values of the tower system in the correction model.

4. Verification of the co-simulation model

The accuracy of the MBD model has already been validated using OpenFAST software [29]. Therefore, this paper does not include a verification of the MBD model. Instead, the dynamic responses of the CS model will be verified against the MBD model.

4.1. Verification of FE model built with OpenSEES solver

The proposed CS model is established based on MBD and FE models. Since the MBD model has already been validated, this section will compare the FE models created using OpenSEES with a more refined model developed in Abaqus through static push-over analysis. This comparison will serve to validate the FE model built with OpenSEES solver.

Fig. 14 illustrates the FE model of the SCHAT developed using Abaqus software. This model employs three-dimensional elements, with embedded truss elements inside the concrete tower acting as reinforcement mesh. Additionally, the model includes rebar elements to simulate prestressed steel reinforcement extending the entire length of the concrete tower from top to bottom. The constitutive behavior of the concrete is represented by a damage plastic model [51], while the steel mesh, tower segments, and prestressed reinforcement are described by bilinear plastic constitutive models. A lumped mass at the top of the model represents the weight of components such as the blades and nacelle, and fixed constraints are applied to both the tower's foundation and the bottom of the prestressed rebar [52]. As depicted in the detailed illustration in Fig. 14, the connection between the steel and concrete towers uses a flange joint, with the top end of the prestressed steel rebar also fixed to this flange.

In the pushover analysis, both Abaqus and OpenSEES models apply loads via displacement. Initially, gravity loads and tendon prestressing

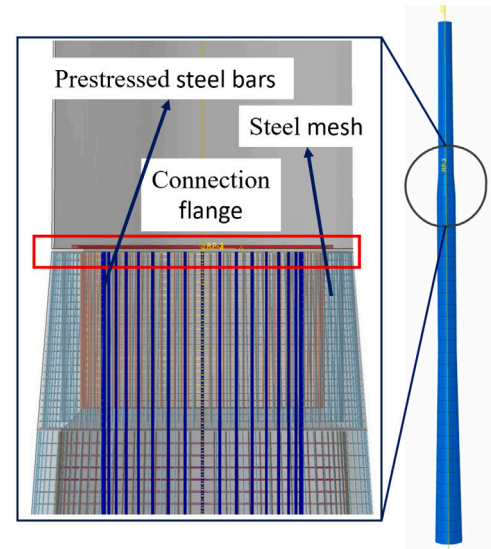


Fig. 14. Model overview and connection detail in Abaqus.

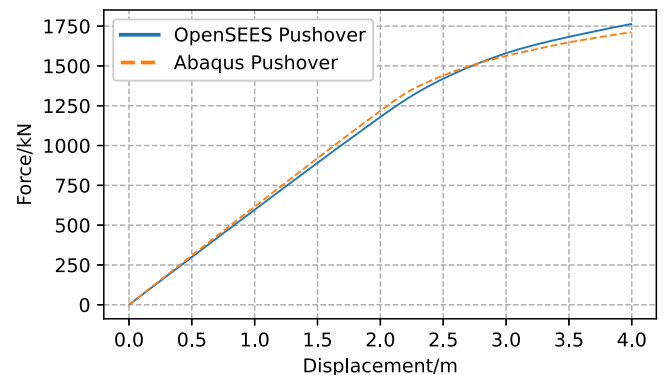


Fig. 15. Comparison of pushover analysis results.

Table 3
Comparison of MBD, LCS and NLCS models.

Model	Blade subsystem	Tower subsystem	Key features
MBD	Modal synthesis method	Modal synthesis method	Fast but limited in handling nonlinear effects.
LCS	Modal synthesis method	Linear fiber beam element	For co-simulation validation.
NLCS	Modal synthesis method	Nonlinear fiber beam element	Highly accurate for nonlinear dynamic behavior.

are applied. Then, a 4 m displacement is applied at the top of the tower in both models to produce displacement–force curves. As shown in Fig. 15, the displacement–force curves from both models are quite similar. The Abaqus model exhibits slightly higher stiffness in the linear stage compared to the OpenSEES model. Despite this minor difference, the results remain very close, validating the effectiveness and accuracy of the OpenSEES model for simulating the structural behavior of the SCHT.

4.2. Verification of transient response

This section compares the dynamic responses of an OWT with SCHT simulated by the MBD model and the proposed CS model under operational conditions. The wind conditions include a hub-height mean wind speed of 12 m/s and a turbulence intensity of 10%.

Table 3 presents a comparison of the Multi-Body Dynamics Model (denoted as “MBD Model”), the Linear Co-Simulation Model (denoted as “LCS Model”) and the Nonlinear Co-Simulation Model (denoted as “NLCS Model”), highlighting their respective modeling techniques and applications. In the MBD Model, both the blade and tower subsystems are modeled using the modal synthesis method, which enables efficient computation for linear response analysis. However, this model is limited in accounting for nonlinear dynamic effects, making it less suitable for scenarios involving complex structural interactions. The LCS Model provides a more refined representation by combining the modal synthesis method for the blade with a linear fiber beam element for the tower. This approach facilitates co-simulation validation and is particularly effective for fast and efficient linear dynamic analyses. In contrast, the NLCS Model achieves higher accuracy by employing a nonlinear fiber beam element for the tower, while still utilizing the modal synthesis method for the blade. This configuration is highly accurate in capturing nonlinear dynamic behavior of SCHT.

As illustrated in Fig. 16, the dynamic responses obtained from the three models are presented under identical operating conditions. The results demonstrate that the MBD and LCS models exhibit nearly identical displacement responses in the tower’s fore-aft direction, indicating a high degree of consistency in their predictive capabilities for this motion. However, the NLCS model shows significantly larger responses in the fore-aft direction due to nonlinear effects. This highlights the great necessity of the NLCS model in capturing nonlinear dynamic behaviors compared to the linear approaches. Figs. 16(b) and 16(c) illustrate the velocity and acceleration responses, where the MBD and LCS models exhibit consistent profiles, demonstrating their agreement under linear dynamic conditions. The frequency response of the NLCS model, as depicted in Fig. 16(d) is also strikingly similar to that of the linear model. Given that the average wind speed $v_0 = 12$ m/s corresponds to high wind speed conditions, it is expected that the NLCS model will enter the nonlinear state. This transition results in significantly larger displacements compared to the LCS model, which is a reasonable and anticipated phenomenon under such conditions.

To provide a comprehensive analysis, this section also compares the time histories of displacement between NLCS and MBD models under low wind speed conditions, characterized by an average wind speed of $v = 2$ m/s. As shown in Fig. 17, at a low wind speed, the displacement is quite small in both directions, indicating that it has not entered the nonlinear state. Under these conditions, the fore-aft displacement of the NLCS model closely agrees with that of the MBD model. In addition, due to the relatively minor nature of side-to-side

Table 4
Wind speed and turbulence intensity.

	Wind speed	Turbulence intensity
Normal	6.48 m/s	0.29%
Extreme	27.7 m/s	0.17%

displacement in these scenarios compared to the displacement in fore-aft direction, the differences between the displacements of the NLCS and MBD models could be considered acceptable.

In summary, the dynamic response comparison between the co-simulation model LCS and the MBD model shows strong consistency, with both models exhibiting nearly identical displacement responses, particularly in the tower’s fore-aft direction. This validates the LCS model’s accuracy in predicting the system’s dynamic behavior under standard conditions, with the MBD model serving as a reliable benchmark. While the LCS model accurately replicates the MBD model’s displacement response, the NLCS model, which accounts for nonlinear effects, shows significantly larger responses, especially in the fore-aft direction, under high wind conditions. This emphasizes the need to consider nonlinearities for accurate predictions under extreme loading scenarios. The frequency spectra of the LCS model closely match those of the MBD model, further confirming the LCS model’s effectiveness in capturing the turbine’s dynamic characteristics. Thus, the LCS model provides a reliable approximation under normal operating conditions, while the NLCS model is essential for more accurate predictions when nonlinear deformations become significant at higher wind speeds.

4.3. Nonlinear dynamic response analysis of OWTs with SCHT under extreme wind conditions

In this section, the dynamic responses of the MBD and NLCS models under various loading conditions will be analyzed. Differences and similarities across different load scenarios will be compared to evaluate the robustness of the results. This comparative analysis aims to highlight the strengths and limitations of these modeling approaches, providing insights into wind turbine tower performance under both typical and extreme conditions. Valuable data support and design guidance for the future development of wind turbines will be also provided through this analysis.

It is well-established that ERA5, the fifth-generation reanalysis of global climate and weather data from the European Centre for Medium-Range Weather Forecasts (ECMWF), offers comprehensive data spanning the past eighty years. According to IEC 61400-1, the extreme wind condition is defined using the 50-year return period wind speed. For this study, data from the ERA5 dataset were collected for a specific location in Tongyu, China, covering the period from January 1, 1973, to September 27, 2023. This 50-year dataset consists of hourly average wind speeds, amounting to a total of 444,762 data points. These data are crucial for calculating the annual average wind speed under typical operational conditions and the maximum average wind speed over a 50-year return period for extreme loading scenarios. Table 4 provides specific values for wind speed and turbulence intensity. Fig. 18 illustrates the hourly average wind speeds at a height of 10 m from 1973 to 2023.

The turbulence standard deviation σ can be determined according to the specification GB/T 18451.1-2022:

$$\sigma = I_{\text{ref}}(0.75V_{\text{hub}} + b) \quad (17)$$

$$b = 5.6 \text{ m/s}$$

where I_{ref} is the reference value for turbulence intensity.

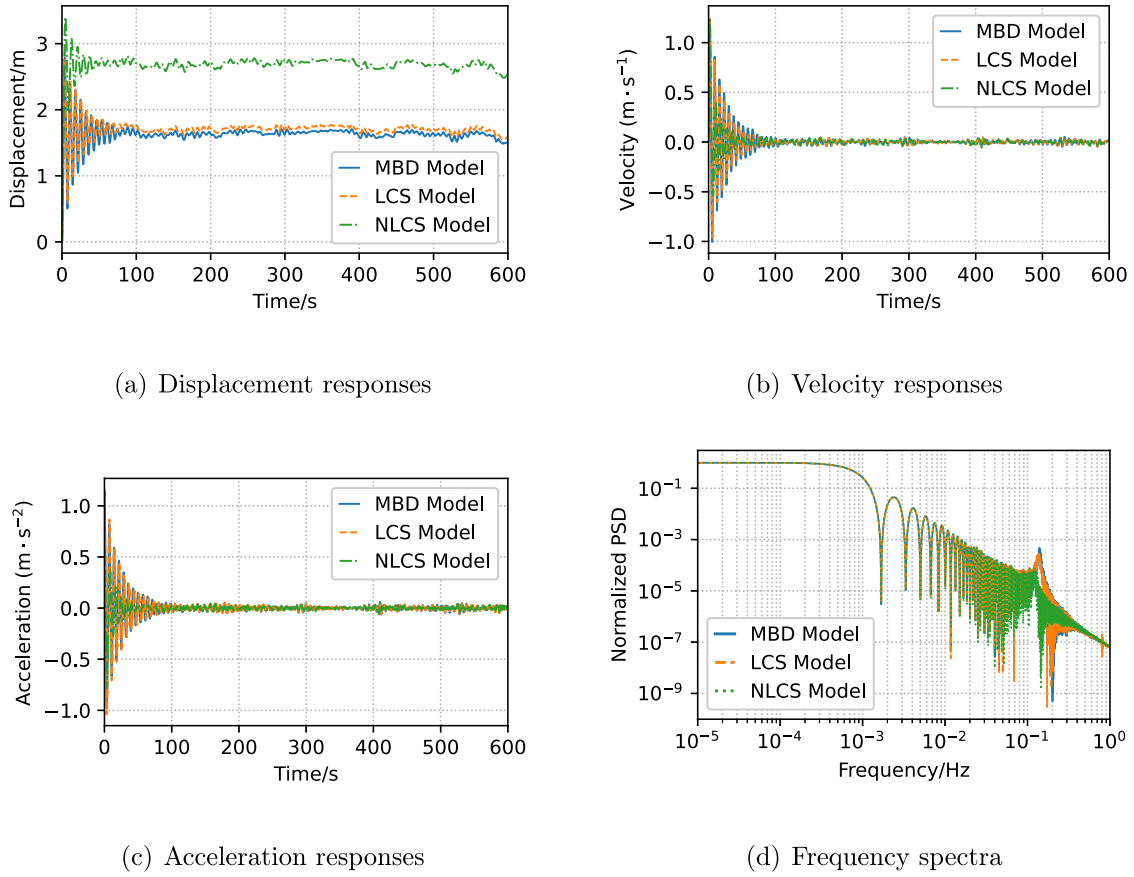


Fig. 16. Comparison of dynamic responses in fore-aft direction.

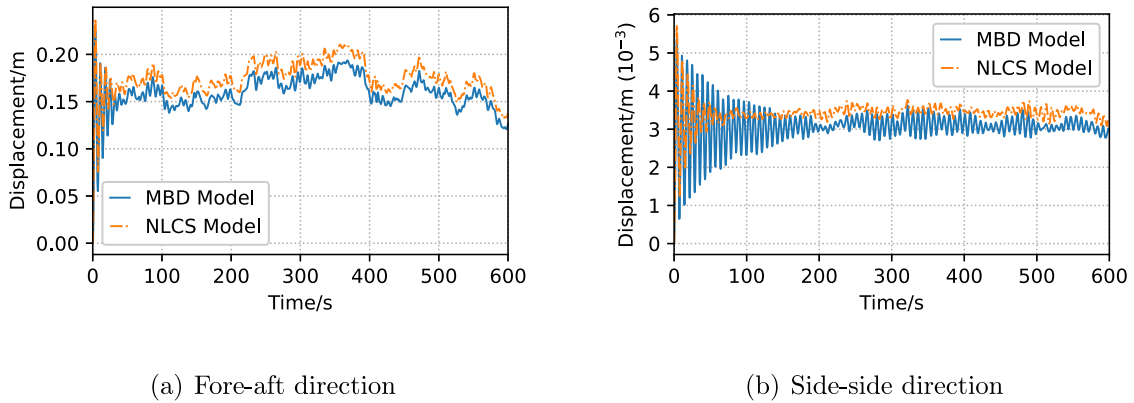


Fig. 17. Comparison of dynamic responses under low wind speed conditions.

Turbulence is classified into four categories, as outlined in GB/T 18451.1-2022. In this paper, the highest class of turbulence characteristics is selected to calculate the reference value for turbulence intensity, denoted as $I_{\text{ref}} = 0.16$. The notation V_{hub} represents the wind speed measured at hub height. Turbulence intensity values are computed using Eq. (18), with the resulting values presented in Table 4 for normal and extreme wind speeds (i.e., the 50-year return period maximum).

$$I = \frac{\sigma}{V_{\text{hub}}} \quad (18)$$

Building upon the previous discussions, we can now proceed with detailed calculations for the wind loads acting on the wind turbine. To capture a comprehensive range of dynamic responses and interactions, the simulation time for wind turbine dynamics should typically be at

least 600 s. During simulations, to avoid the influence of transient responses—which can introduce significant outliers and anomalies in the data—the initial dynamic response when the wind turbine begins operation is usually omitted from the analysis. In the case of linear models, the transient response that occurs during the aerodynamic phase of the wind turbine can be disregarded, as it has negligible impact on the subsequent steady-state dynamics. However, nonlinear models present a more complex scenario. In these systems, large transient responses could lead to phenomena such as plastic deformation or even structural failure. These responses must be carefully considered, as they significantly influence the later stages of simulation and overall turbine behavior. In this study, the dynamic time history simulation of wind turbines will involve a controlled linear start-up process. The rotor speed will be methodically accelerated from 0 to the required

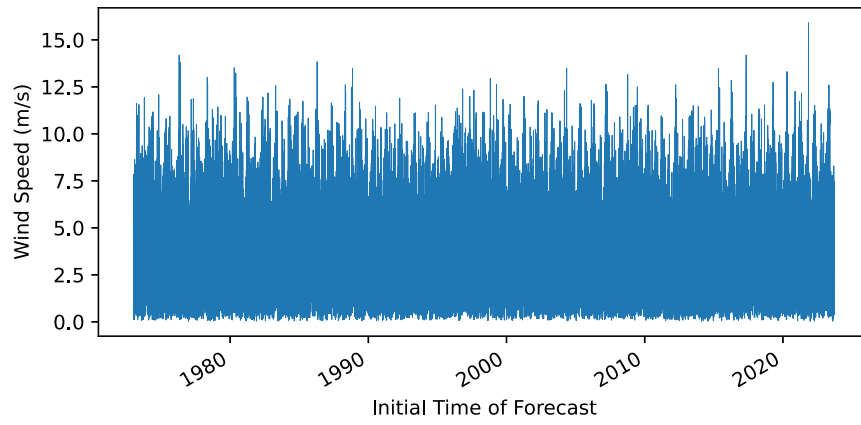


Fig. 18. Wind speed from 1973 to 2023.

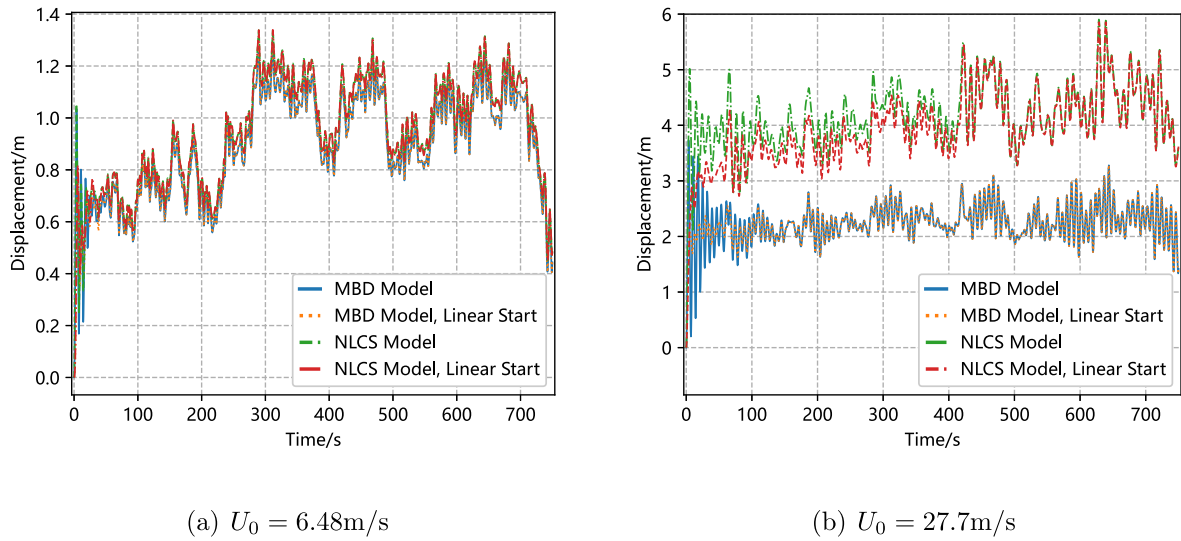


Fig. 19. Comparison of start-up process in fore-aft direction.

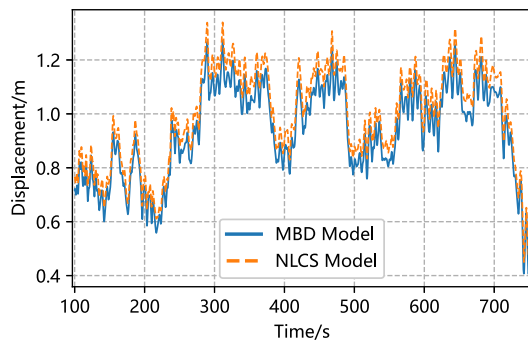
operational rotational speed within 5 s. After this initial phase, the analysis will focus exclusively on data collected after the first 100 s of operation. This approach ensures that the data reflects the turbine's performance under stable operating conditions, providing a more accurate representation of its dynamic behavior under various wind load scenarios.

As illustrated in Fig. 19, both linear and nonlinear models exhibit the elimination of initial transient responses when the blade is started linearly, with no subsequent impact on the simulation. This comparison highlights that employing a controlled linear start-up method can effectively mitigate transient response effects on wind turbine structures. It allows the nonlinear model to transition more smoothly and promptly to the nonlinear stage during simulation.

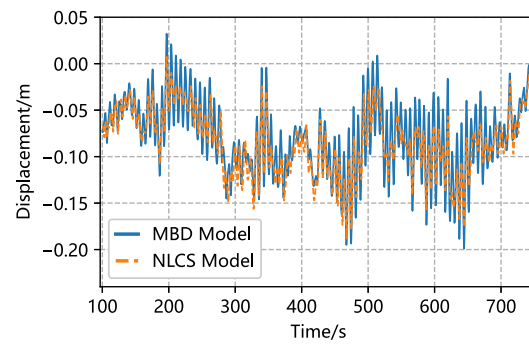
The top displacements of the tower under two wind conditions are shown in Figs. 20 and 21. For normal wind speed, the results from the MBD model are close to those of the NLCS model. Specifically, the maximum fore-aft displacement observed in the MBD model is 1.27 m, while the side-side displacement is 0.2 m. Comparatively, the NLCS model exhibits a slightly larger maximum fore-aft displacement of 1.34 m and a side-side displacement of 0.19 m. This close agreement between the MBD and NLCS models under normal wind conditions highlights the robustness of the NLCS model in simulating the dynamic response of OWTs with SHT. The slight discrepancy in fore-aft displacement may be attributed to the inherent nonlinearities accounted for the NLCS model, which makes it slightly more sensitive to dynamic variations in wind loads.

In extreme wind speed conditions, as shown in Fig. 21, the differences between the MBD model and the NLCS model become more pronounced. The MBD model indicates a maximum fore-aft displacement of 2.47 m and a maximum side-side displacement of 0.36 m. In contrast, the NLCS model shows substantially higher values, with the maximum fore-aft displacement reaching 5.90 m and the maximum side-side displacement reaching 0.62 m. This significant discrepancy under extreme wind conditions highlights the limitations of the MBD model in accurately capturing the dynamic behavior and flexibility of OWTs with SHT under severe loading scenarios. The NLCS model demonstrates its ability to account for nonlinear interactions and more complex dynamics, reflecting a higher sensitivity to extreme wind forces.

The considerable differences in displacement values suggest that for design and safety evaluations under extreme wind conditions, the NLCS model is more reliable. It can provide critical insights into the potential structural challenges and failure mechanisms that may arise, ensuring that wind turbine designs are robust enough to withstand such extreme events. Overall, while the MBD model may be sufficient for normal operational conditions, its limitations are apparent under extreme wind scenarios. The NLCS model offers a more accurate representation of the wind turbine's response, making it essential for accurately predicting performance and ensuring structural integrity in the face of extreme wind conditions.

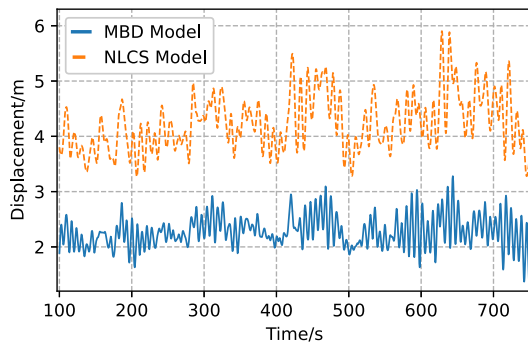


(a) Fore-aft direction

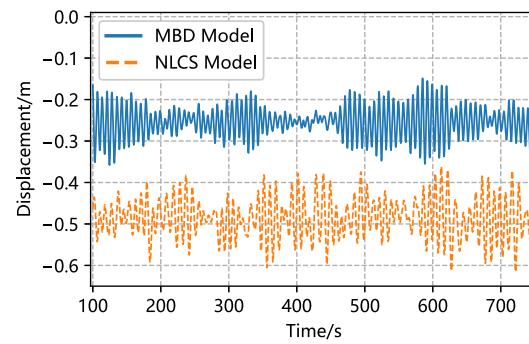


(b) Side-side direction

Fig. 20. Comparison of dynamic responses under normal wind speed conditions.



(a) Fore-aft direction



(b) Side-side direction

Fig. 21. Comparison of dynamic responses under extreme wind speed conditions.

5. Conclusions

In this paper, a co-simulation method has been developed to investigate the nonlinear dynamic response of OWTs with SCHT. This approach involves breaking down the SCHT wind turbine into blade and tower subsystems using a force–force coupling method. Specifically, the equations of motion for the blade system have been derived based on the Euler–Lagrange equations, establishing a multi-body dynamic model. The finite element method has been employed to solve the structural dynamics of the SCHT, utilizing OpenSEES software. To account for the interactions between these two separate systems, an iterative coupling method has been applied during simultaneous solution of the subsystems. A numerical example has been studied to validate the proposed co-simulation method. The following conclusions have been reached:

1. Accuracy of the Co-Simulation Method: The proposed co-simulation method accurately captures the dynamic responses of OWTs with SCHT. Comparisons with established MBD and refined FE models demonstrate that the co-simulation approach yields reliable results under both linear and nonlinear conditions.

2. Effectiveness of Force–Force Coupling: The force–force coupling method effectively decouples the blade and tower subsystems, allowing each to be solved independently while still accounting for their interactions. This approach simplifies the computational process without sacrificing accuracy.

3. Nonlinear Dynamic Behavior: The NLCS model effectively captures the nonlinear dynamic behavior of the system, especially under high wind speed conditions. The ability to transition smoothly into the nonlinear state and accurately predict larger displacements highlights the model's robustness.

The co-simulation method provides a valuable tool for analyzing and designing OWTs with SCHT. Future work could explore further refinements to enhance computational efficiency and extend the method to other types of wind turbine systems and loading scenarios.

CRediT authorship contribution statement

Jun Xu: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zeyu He:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation. **Dan Wang:** Writing – review & editing, Writing – original draft, Validation. **Guangling He:** Writing – review & editing, Supervision, Resources, Project administration. **Qiang Wu:** Writing – review & editing, Resources, Project administration. **Zili Zhang:** Writing – review & editing, Writing – original draft, Supervision.

Replication of results

Some or all data, models, or code generated or used during the study are available from the corresponding author by request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Mathematical model of the 8-DOF SCHT

A.1. System kinetic energy

The dynamic equations for an 8-DOF SCHT wind turbine generator can be obtained through the application of the Euler–Lagrange equation:

$$\frac{d}{dt} \frac{\partial T(t, \mathbf{q}(t), \dot{\mathbf{q}}(t))}{\partial \dot{q}_i(t)} - \frac{\partial T(t, \mathbf{q}(t), \dot{\mathbf{q}}(t))}{\partial q_i(t)} + \frac{\partial V(t, \mathbf{q}(t))}{\partial q_i(t)} = Q_i(t) \quad (\text{A.1})$$

where T and V represent the kinetic and potential energy of the system, respectively. The vector $\mathbf{q}(t) = [q_1, q_2, \dots, q_8]^T$ denotes the generalized coordinates, and $Q_i(t)$ is the generalized force acting on the i th degree of freedom in $\mathbf{q}(t)$.

As shown in Fig. 2, the absolute displacements and velocities of the nacelle (“nac”) in the fore-aft (“fa”) and side-side (“ss”) directions can be expressed as follows:

$$u_{\text{nac}}^{\text{fa}} = q_7, \quad u_{\text{nac}}^{\text{ss}} = q_8; \quad v_{\text{nac}}^{\text{fa}} = \dot{q}_7, \quad v_{\text{nac}}^{\text{ss}} = \dot{q}_8 \quad (\text{A.2})$$

Hence, the absolute velocity of the nacelle is:

$$v_{\text{nac}} = \sqrt{(v_{\text{nac}}^{\text{fa}})^2 + (v_{\text{nac}}^{\text{ss}})^2} \quad (\text{A.3})$$

Additionally, by defining a local coordinate system $o'x'y'z'$ as the origin, the following considerations are made. For the j th blade, at a distance r relative to the hub, the edgewise displacement $u_{j,e}(r, t)$ and flapwise displacement $u_{j,f}(r, t)$ are defined. The relative displacement of a specific blade element on the blade can be expressed as:

$$u_{j,e}(r, t) = q_j \cdot \phi_{1e}(r), \quad j = 1, 2, 3 \quad (\text{A.4})$$

$$u_{j,f}(r, t) = q_j \cdot \phi_{1f}(r), \quad j = 4, 5, 6$$

where $\phi_{1e}(r)$ and $\phi_{1f}(r)$ denote the first modal shapes for edgewise and flapwise motions respectively.

Hence, the relative displacement per unit length of the j th blade with respect to the local coordinate system is given by:

$$\begin{aligned} x'_r &= u_{\text{nac}}^{\text{ss}} + r \cdot \sin \psi_j + u_{e,j} \cos \psi_j \\ y'_r &= u_{\text{nac}}^{\text{fa}} + u_{f,j} \\ z'_r &= r \cdot \cos \psi_j - u_{e,j} \sin \psi_j \end{aligned} \quad (\text{A.5})$$

Likewise, the first derivative of the relative displacement can be expressed as:

$$\begin{aligned} \dot{x}'_r &= v_{\text{nac}}^{\text{ss}} - \Omega r \cdot \cos \psi_j + \dot{q}_j \cdot \phi_{1e}(r) \cos \psi_j - q_j \Omega \cdot \phi_{1e}(r) \sin \psi_j \\ \dot{y}'_r &= v_{\text{nac}}^{\text{fa}} + \dot{q}_{j+3} \cdot \phi_{1f}(r) \\ \dot{z}'_r &= -\Omega r \cdot \sin \psi_j - \dot{q}_j \cdot \phi_{1e}(r) \sin \psi_j - q_j \Omega \cdot \phi_{1e}(r) \cos \psi_j \end{aligned} \quad (\text{A.6})$$

As a result, the absolute velocity per unit length of the blade with respect to the local coordinate system is given by:

$$v_{bj}(r, t) = \sqrt{(\dot{x}'_r)^2 + (\dot{y}'_r)^2 + (\dot{z}'_r)^2} \quad (\text{A.7})$$

Likewise, for a unit length of the tower, the absolute velocity is given by:

$$v_{\text{tow}} = \sqrt{(\dot{q}_7 \cdot \phi_{1t}(r))^2 + (\dot{q}_8 \cdot \phi_{1t}(r))^2} \quad (\text{A.8})$$

Therefore, the total system kinetic energy can be expressed as:

$$T = \frac{1}{2} \sum_{j=1}^3 \left[\int_0^R \bar{m} \cdot v_{bj}^2(r, t) dr + \frac{1}{2} M_{\text{nac}} \cdot v_{\text{nac}}^2 \right] + \frac{1}{2} \int_0^{H_{\text{tow}}} \bar{m}_{\text{tow}} \cdot v_{\text{tow}}^2 dz \quad (\text{A.9})$$

where $\bar{m}$ represents the mass per unit length of the blade, M_{nac} is the total mass of the nacelle, $\bar{m}_{\text{tow}}$ is the mass per unit length of the tower and R , H_{tow} denote the blade's radius and tower's height, respectively.

A.2. System potential energy

The total potential energy of the blades encompasses the strain potential energy, the potential energy arising from centrifugal motion, and the gravitational potential energy term. The collective potential energy of the three-blade system can be expressed as follows [53]:

$$\begin{aligned} V_b = \frac{1}{2} \sum_{j=1}^3 & \left[(k_{\text{eg}} + k_{\text{ge, eg}} - k_{\text{gr, eg}} \cdot \cos \psi_j) \cdot q_j^2 \right. \\ & \left. + (k_{\text{fg}} + k_{\text{ge, fp}} - k_{\text{gr, fp}} \cdot \cos \psi_j) \cdot q_{j+3}^2 \right] \end{aligned} \quad (\text{A.10})$$

where:

$$\begin{aligned} k_{\text{eg}} &= \int_0^R EI_{\text{eg}}(r)(\phi'_{1e})^2 dr, & k_{\text{fg}} &= \int_0^R EI_{\text{fg}}(r)(\phi'_{1f})^2 dr \\ k_{\text{ge, eg}} &= \Omega \int_0^R \int_r^R [\bar{m}(\xi) \xi d\xi] (\phi'_{1e})^2 dr, & k_{\text{ge, fp}} &= \Omega \int_0^R \int_r^R [\bar{m}(\xi) \xi d\xi] (\phi'_{1f})^2 dr \\ k_{\text{gr, eg}} &= g \int_0^R \int_r^R [\bar{m}(\xi) d\xi] (\phi'_{1e})^2 dr, & k_{\text{gr, fp}} &= g \int_0^R \int_r^R [\bar{m}(\xi) d\xi] (\phi'_{1f})^2 dr \end{aligned} \quad (\text{A.11})$$

where I_f and I_e represent the moments of inertia in the flapwise and edgewise directions, respectively. g is the gravitational acceleration, E represents the elastic modulus of blades, and $()'$ and $()''$ denote the first and second derivatives with respect to the distance r .

Consequently, the complete potential energy of the onshore wind turbine system can be expressed as:

$$V = V_b + \frac{1}{2} k_t^{\text{fa}} \cdot q_7^2 + \frac{1}{2} k_t^{\text{ss}} \cdot q_8^2 \quad (\text{A.12})$$

where k_t^{fa} , k_t^{ss} represent the stiffness of the tower in the fore-aft and side-side directions respectively. It is assumed that the stiffness in both directions is consistent and can be derived from the following equation:

$$k_t^{\text{fa}} = k_t^{\text{ss}} = \int_0^{H_c} \bar{E}_c I_t(z)(\phi'_{1t})^2 dz + \int_{H_c}^{H_{\text{tow}}} E_s I_t(z)(\phi'_{1t})^2 dz \quad (\text{A.13})$$

where H_c represents the height of the concrete tower section, ϕ_{1t} signifies the first modal shape of the tower motion, and E_s represents the elastic modulus of the steel tower. In practical numerical computations, the tower can be discretized into multiple segments for analysis. The moment of inertia of the annular cross-section for steel tower segments can be directly calculated. However, for a composite cross-section of a reinforced concrete tower segment, the bending stiffness of the concrete and the rebars should be computed separately and then superimposed [54], as shown in Eq. (A.14).

$$\bar{E}_c I_t(z) = E_c I_c(z) + E_{sb} I_{sb}(z) \quad (\text{A.14})$$

where E_c and E_{sb} represent the elastic modulus of concrete and rebar, respectively, while $I_c(z)$ and $I_{sb}(z)$ correspond to the moments of inertia of the concrete tower section and steel tower section.

By substituting Eqs. (A.9) and (A.12) into Eq. (A.1), the governing equations of motion for the system can be derived as follows:

$$\tilde{\mathbf{M}}\ddot{\mathbf{q}} + \tilde{\mathbf{C}}\dot{\mathbf{q}} + \tilde{\mathbf{K}}\mathbf{q} = \tilde{\mathbf{Q}}_{\text{wind}} \quad (\text{A.15})$$

where $\tilde{\mathbf{M}}$, $\tilde{\mathbf{C}}$, $\tilde{\mathbf{K}}$ are the mass, damping, and stiffness matrices of the system, respectively. The terms $\tilde{\mathbf{Q}}_{\text{wind}}$ is the generalized force vectors of the system due to wind loads.

A.3. System matrix of the 8-DOF SCHT

The matrix representation of the 8-DOF SCHT system is expressed as follows:

$$\mathbf{M} = \begin{bmatrix} m_{1e} & 0 & 0 & 0 & 0 & 0 & 0 & m_{18} \\ 0 & m_{1e} & 0 & 0 & 0 & 0 & 0 & m_{18} \\ 0 & 0 & m_{1e} & 0 & 0 & 0 & m_{47} & m_{18} \\ 0 & 0 & 0 & m_{1f} & 0 & 0 & m_{47} & 0 \\ 0 & 0 & 0 & 0 & m_{1f} & 0 & m_{47} & 0 \\ 0 & 0 & 0 & 0 & 0 & m_{1f} & m_{47} & 0 \\ 0 & 0 & 0 & m_{47} & m_{47} & m_{47} & m_7 & 0 \\ m_{18} & m_{38} & m_{38} & 0 & 0 & 0 & 0 & m_8 \end{bmatrix} \quad (\text{A.16})$$

where:

$$\begin{aligned} m_{1e} &= \int_0^R \bar{m}(r) \phi_{1e}^2 dr, & m_{1f} &= \int_0^R \bar{m}(r) \phi_{1f}^2 dr, \\ m_{18} &= \int_0^R \bar{m}(r) \phi_{1e} \phi_{1f} dr \cdot \cos \psi_j, & j &= 1, 2, 3 & m_{47} &= \int_0^R \bar{m}(r) \phi_{1f} dr, \\ m_7 &= m_8 = 3m_0 + M_{nac} + M_{hub} + \int_0^{H_{tow}} \bar{M}(z) \phi_{1t}^2 dz \end{aligned} \quad (\text{A.17})$$

Parameters M_{nac} and M_{hub} denote the masses of the nacelle and hub, respectively. m_0 is the total mass of each blade, and $\bar{m}(r)$ is the mass per unit length of the blade. ϕ_{1e} and ϕ_{1f} are the first mode shapes of the edgewise and flapwise directions of the blade, respectively. $\bar{M}(z)$ represents the mass per unit length of the tower. ϕ_{1t} is the first mode shape of the tower. ψ_j is the angle between the x -axis and the j th blade. R is the radius of the rotor. H_{tow} is the height of the tower.

$$\mathbf{K} = \begin{bmatrix} k_{b1,e} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & k_{b2,e} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & k_{b3,e} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k_{b1,f} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & k_{b2,f} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & k_{b3,f} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & k_7 & 0 \\ -\Omega_{18}^2 & -\Omega_{28}^2 & -\Omega_{38}^2 & 0 & 0 & 0 & 0 & k_8 \end{bmatrix} \quad (\text{A.18})$$

where:

$$\begin{aligned} k_{b1,e} &= k_{eg} + k_{ge,eg} - k_{gr,eg} \cos \psi_j - \Omega^2 m_{1e}, & k_{b1,f} &= k_{fg} + k_{ge,fp} \\ &- k_{gr,fp} \cos \psi_j, \end{aligned} \quad (\text{A.19})$$

$$k_7 = k_8 = \int_0^{H_c} \bar{E}_c I_t (\phi_{1t}'')^2 dz + \int_{H_c}^{H_{tow}} E_s I_t (\phi_{1t}'')^2 dz$$

and:

$$\begin{aligned} k_{eg} &= \int_0^R E I_{eg}(r) (\phi_{1e}'')^2 dr, & k_{fg} &= \int_0^R E I_{fg}(r) (\phi_{1f}'')^2 dr \\ k_{ge,eg} &= \Omega \int_0^R \int_r^R [\bar{m}(\xi) \xi d\xi] (\phi_{1e}')^2 dr, & k_{ge,fp} &= \Omega \int_0^R \int_r^R [\bar{m}(\xi) \xi d\xi] (\phi_{1f}')^2 dr \\ k_{gr,eg} &= g \int_0^R \int_r^R [\bar{m}(\xi) d\xi] (\phi_{1e}')^2 dr, & k_{gr,fp} &= g \int_0^R \int_r^R [\bar{m}(\xi) d\xi] (\phi_{1f}')^2 dr \end{aligned} \quad (\text{A.20})$$

Parameters I_f and I_e represent the moments of inertia in the flapwise and edgewise directions, respectively. g is the gravitational acceleration, E represents the elastic modulus of blades, and $()'$ and $()''$ denote the first and second derivatives with respect to the distance r . I_t is the moment of inertia of the tower cross-section. H_c is the height of the reinforced concrete tower. E_s and E_c are the elastic moduli of

the steel mesh and concrete, respectively.

$$\mathbf{C} = \begin{bmatrix} c_{b,e} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & c_{b,e} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & c_{b,e} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{b,f} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{b,f} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{b,f} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & c_7 & 0 \\ -2\Omega m_{81} & -2\Omega m_{82} & -2\Omega m_{83} & 0 & 0 & 0 & 0 & c_8 \end{bmatrix} \quad (\text{A.21})$$

where:

$$\bar{m}_{8j} = \int_0^R \bar{m}(r) \phi_{1e} dr \cdot \sin \psi_j, \quad j = 1, 2, 3 \quad (\text{A.22})$$

Parameters $c_{b,e}$ and $c_{b,f}$ represent the damping in the edgewise and flapwise directions of the blades, while c_7 and c_8 denote the damping in the fore-aft and side-side directions of the tower, respectively.

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